ENTERPRISE BUSINESS INTELLIGENCE SUITE

Multi-Region Business Intelligence and Analytics

Project: Enterprise Business Intelligence Suite for a Multi-Region Company
Specialization: Data Analytics
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Technology: Microsoft Power BI, Power Query, DAX and Microsoft Excel

1. EXECUTIVE SUMMARY

The Enterprise Business Intelligence Suite is a data analytics project designed to provide a consolidated view of business performance across sales, customers, products, markets, regions and supply chain operations.

The project uses the DataCo SMART SUPPLY CHAIN dataset to develop an interactive Business Intelligence solution using Microsoft Power BI.

The solution involves importing and preparing data, creating a star-schema data model, developing DAX measures, designing interactive dashboards and implementing Row-Level Security (RLS) to restrict regional data access.

Four dashboards were developed:

1. Executive Overview
2. Sales Performance Analysis
3. Supply Chain & Delivery Analysis
4. Customer & Market Insights

The project demonstrates how business data can be transformed into meaningful reports that support data-driven decision-making.

2. PROJECT INTRODUCTION

Organizations operating across multiple regions generate large amounts of sales, customer, product and operational data.

Analyzing these datasets separately can make it difficult to understand overall business performance, identify trends and monitor operational efficiency.

The objective of this project is to develop a centralized Business Intelligence reporting solution that organizes available business data into an interactive Power BI environment.

The solution enables users to analyze business performance through KPIs, charts, filters and regional security restrictions.

2.1 Project Objectives

- Consolidate available business data into a centralized reporting model.
- Clean and transform raw data using Power Query.
- Design a star-schema data model.
- Develop reusable DAX measures for business KPIs.
- Build interactive dashboards for different business areas.
- Analyze sales, customer, product and supply chain performance.
- Implement Row-Level Security for regional data access.
- Establish a repeatable data refresh process.
- Present meaningful business information through visualizations.

3. TOOLS AND TECHNOLOGIES

The following tools and technologies were used.

Technology	Purpose
Microsoft Power BI Desktop	Data modelling, visualization and dashboard development
Power Query Editor	Data cleaning and transformation
Microsoft Excel	Source data storage
DAX	KPI calculations and analytical measures
Star Schema	Organizing fact and dimension tables
Row-Level Security	Restricting regional data access

4. DATASET DESCRIPTION

4.1 Dataset Source

Dataset Name: DataCo SMART SUPPLY CHAIN Dataset

The dataset contains information related to sales transactions, orders, customers, products, markets, regions and shipping operations.

The original dataset was provided in CSV format and converted into an Excel workbook for Power BI integration.

The working dataset contained approximately 180,520 records.

4.2 Key Data Attributes

The dataset includes information related to:

Sales and Orders

- Order ID
- Sales
- Order date
- Order quantity
- Order status

Customer Information

- Customer ID
- Customer segment
- Customer location

Product Information

- Product ID
- Product name
- Product category

Geographical Information

- Market
- Order region
- Customer region

Shipping and Delivery

- Shipping mode
- Actual shipping days
- Scheduled shipping days
- Delivery status
- Late delivery risk

4.3 Business Areas Covered

The available dataset supports analysis of:

- Sales performance
- Customer purchasing behaviour
- Product performance
- Market and regional performance
- Shipping and delivery operations

5. DATA CLEANING AND TRANSFORMATION

Data preparation was performed using Power Query Editor in Power BI Desktop.

The objective was to improve data consistency, prepare suitable data types and organize the data for analytical modelling.

5.1 Data Import

The converted Excel workbook was imported into Power BI Desktop.

The imported data was reviewed to understand the available columns, data types and structure.

5.2 Data Cleaning Activities

1. Reviewed column names and data types.
2. Inspected missing values and column quality.
3. Removed unnecessary columns that were not required for analysis.
4. Reviewed duplicate records.
5. Corrected date and date-time data types.
6. Prepared customer, product, region and date dimension tables.
7. Organized the data into suitable fact and dimension tables.
8. Disabled loading for the original staging query where appropriate to avoid duplicate model tables and ambiguous relationships.

5.3 Date Transformation

The order date column initially contained date and time information.

The column was converted into a consistent date-only format to support date-based relationships and reporting.

A separate Date Dimension was used for year and month analysis.

5.4 Data Loading

After cleaning and transformation, the prepared tables were loaded into the Power BI data model.

6. DATA MODELLING – STAR SCHEMA

A star-schema data model was designed to organize transactional data and descriptive information.

The model consists of one central fact table supported by dimension tables.

6.1 Fact Table

FactSales

The FactSales table contains transactional sales and order information.

Important attributes include:

- Order ID
- Customer ID
- Product ID
- Order Date
- Sales
- Order Quantity
- Shipping Information
- Delivery Risk
- Region

6.2 Dimension Tables

DimCustomer

Contains customer-related information used for customer analysis and segmentation.

DimProduct

Contains product-related attributes such as product names and product categories.

DimDate

Contains calendar information such as:

- Date
- Year
- Month
- Month Name

DimRegion

Contains geographical and regional information used for regional analysis.

6.3 Relationship Design

The dimension tables are connected to the central FactSales table.

Dimension Table	FactSales Matching Field
DimCustomer	Customer ID
DimProduct	Product ID
DimDate	Order Date
DimRegion	Corresponding region key or matching region field

The intended relationship structure is one-to-many, with dimension tables on the one side and FactSales on the many side.

This model allows users to analyze transactional data using descriptive attributes from the dimension tables.

7. DAX MEASURES AND KPI DEVELOPMENT

DAX measures were created to calculate important business metrics.

The measures are reusable across different dashboard pages.

7.1 Total Sales

Total Sales =

SUM(FactSales[Sales])

Calculates total sales revenue.

7.2 Total Profit

Total Profit =

SUM(FactSales[Order Profit Per Order])

Calculates the sum of the order-profit field.

Note: This calculation must be interpreted according to the source data grain. If the value repeats across multiple item rows, additional validation is required to avoid double counting.

7.3 Total Orders

```
Total Orders =

DISTINCTCOUNT(FactSales[Order Id])
```

Calculates the total number of unique orders.

7.4 Total Quantity Sold

```
Total Quantity Sold =

SUM(FactSales[Order Item Quantity])
```

Calculates the total quantity of products ordered.

7.5 Late Delivery Orders

```
Late Delivery Orders =

CALCULATE(
    DISTINCTCOUNT(FactSales[Order Id]),
    FactSales[Late_delivery_risk] = 1
)
```

Calculates the number of distinct orders flagged with late-delivery risk.

7.6 Total Customers

```
Total Customers =

DISTINCTCOUNT(FactSales[CustomerID])
```

Calculates the number of unique customers.

7.7 Average Order Value

```
Average Order Value =

DIVIDE(
    [Total Sales],
    [Total Orders],
    0
)
```

Calculates average sales per order.

7.8 Late Delivery Risk Percentage

```
Late Delivery Risk % =
DIVIDE(
    [Late Delivery Orders],
    [Total Orders],
    0
)
```

Calculates the percentage of orders flagged with late-delivery risk.

The measure was formatted as a percentage in Power BI.

Important: Late Delivery Risk % represents the proportion of orders flagged as at risk. It should not automatically be interpreted as the actual late-delivery percentage.

7.9 Average Actual Shipping Days

```
Average Actual Shipping Days =
AVERAGE(FactSales[Days for shipping (real)])
```

Calculates average actual shipping duration, provided the corresponding field is available and numeric.

8. DASHBOARD DEVELOPMENT

Four interactive dashboards were developed to provide different analytical perspectives.

8.1 DASHBOARD 1 – EXECUTIVE OVERVIEW

Title: ENTERPRISE BUSINESS INTELLIGENCE

Subtitle: Executive Overview | Sales & Supply Chain

Objective

To provide a high-level overview of business performance through important KPIs and visualizations.

KPI Cards

1. Total Sales
2. Total Profit
3. Total Orders
4. Late Delivery Orders

Visualizations

1. Monthly Sales Trend – Displays sales performance across months.
2. Sales by Region – Compares sales across geographical regions.
3. Sales by Product Category – Identifies product categories contributing to sales.
4. Orders by Shipping Mode – Displays order distribution across shipping methods.

Slicers

- Year
- Market
- Region

Business Purpose

The Executive Overview provides a consolidated view of sales and operational performance, helping users monitor important business indicators.

8.2 DASHBOARD 2 – SALES PERFORMANCE ANALYSIS

Title: SALES PERFORMANCE ANALYSIS

Subtitle: Revenue Trends | Product Performance | Market Insights

Objective

To analyze sales performance across products, markets and time periods.

KPI Cards

1. Total Sales
2. Total Profit
3. Total Orders
4. Average Order Value

Visualizations

1. Monthly Sales Trend
2. Sales by Product Category
3. Top 10 Products by Sales
4. Sales by Market

Slicers

- Year
- Market
- Product Category

Business Purpose

This dashboard helps identify high-performing products, understand market contributions and monitor sales trends.

8.3 DASHBOARD 3 – SUPPLY CHAIN & DELIVERY ANALYSIS

Title: SUPPLY CHAIN & DELIVERY ANALYSIS

Subtitle: Delivery Performance | Shipping Efficiency | Operational Risk

Objective

To analyze shipping operations, delivery performance and operational risk indicators.

KPI Cards

1. Total Orders
2. Late Delivery Orders
3. Average Actual Shipping Days
4. Late Delivery Risk %

Visualizations

1. Late Deliveries by Region
2. Orders by Shipping Mode
3. Orders by Delivery Status
4. Actual vs Scheduled Shipping Days

Slicers

- Year
- Market
- Region

Business Purpose

This dashboard helps monitor shipping activity, compare delivery durations and identify regions with higher late-delivery risk.

8.4 DASHBOARD 4 – CUSTOMER & MARKET INSIGHTS

Title: CUSTOMER & MARKET INSIGHTS

Subtitle: Customer Segmentation | Purchasing Behavior | Regional Performance

Objective

To understand customer purchasing behaviour and analyze sales contributions across customer segments and regions.

KPI Cards

1. Total Customers
2. Total Orders
3. Total Sales
4. Average Order Value

Visualizations

1. Sales by Customer Segment
2. Top 10 Customers by Sales
3. Orders by Market
4. Customer Count by Region

Slicers

- Year
- Market
- Customer Segment

Business Purpose

This dashboard helps identify important customer segments, understand purchasing patterns and compare customer contributions across regions.

9. ROW-LEVEL SECURITY (RLS)

9.1 Introduction

Row-Level Security is a Power BI security feature that restricts data access based on defined conditions.

It allows different users to view only the records they are authorized to access.

For this project, regional security roles were implemented to demonstrate controlled access to regional sales information.

9.2 Objective

The objective of RLS is to ensure that regional managers can access only their respective regional data while using the same Power BI report.

9.3 Roles Created

Two static security roles were created.

Role Name	Region Access
RegionalManager	South Asia
SoutheastAsiaManager	Southeast Asia

9.4 Security Rules

RegionalManager

[Order Region] = "South Asia"

SoutheastAsiaManager

[Order Region] = "Southeast Asia"

These filters were applied directly to the FactSales table.

9.5 Testing Procedure

The roles were tested using the View As feature in Power BI Desktop.

Steps:

1. Open the Modeling tab.
2. Select View As.
3. Choose the required security role.
4. Apply the role.
5. Review the dashboard pages.
6. Verify that only the authorized region's records contribute to the report.

9.6 Result

Both regional roles were created and tested successfully.

The report demonstrated how the same dashboards can display different regional results depending on the selected security role.

9.7 Future Enhancement

Dynamic RLS can be implemented using a user-to-region mapping table and the USERPRINCIPALNAME() function.

For actual user access, the report must be published to Power BI Service and users must be assigned to the appropriate roles.

10. DATA REFRESH WORKFLOW

10.1 Introduction

Data refresh ensures that the Power BI report reflects the latest information available in the source workbook.

The project uses a locally stored Excel workbook as its data source.

10.2 Data Flow

Excel Workbook → Power Query → Data Model → DAX Measures → Power BI Dashboards

10.3 Refresh Method

Manual refresh was implemented using Power BI Desktop.

Refresh Procedure

1. Update the source Excel workbook when new data is available.
2. Save the workbook.
3. Open the Power BI Desktop project.
4. Select Home → Refresh.
5. Power BI retrieves the updated source data.
6. Power Query transformations are executed again.
7. The data model is updated.
8. Dashboard visuals reflect the refreshed data.

10.4 Benefits

- Simple and easy to maintain.
- No additional cloud infrastructure required.
- Existing Power Query transformations are reusable.
- Suitable for a local internship project demonstration.

10.5 Limitations

Automated scheduled refresh using Power BI Service, Azure Data Factory or scheduled SQL jobs was not implemented.

The current solution uses manual refresh in Power BI Desktop.

Automated refresh can be considered as a future enhancement.

11. DATA VALIDATION AND TESTING

Data validation was performed to check the reliability of the Power BI model and dashboard calculations.

11.1 KPI Validation

Important measures were checked against the source data.

The validation covered:

- Total Sales
- Total Orders
- Total Customers
- Total Quantity Sold
- Late Delivery Orders
- Average Order Value
- Late Delivery Risk Percentage

The Total Profit calculation requires particular attention to the source data grain to avoid double counting.

11.2 Relationship Validation

The fact and dimension table relationships were reviewed.

The validation focused on:

- Active relationships.
- Correct matching columns.
- Unique dimension keys.
- Appropriate filter directions.
- Avoiding ambiguous relationship paths.

11.3 Dashboard Validation

All four dashboards were reviewed to ensure that:

- KPI cards display the intended measures.
- Charts use the correct fields.
- Slicers filter the relevant visuals.
- Visual titles and labels are understandable.
- No unintended blank or incorrect results are displayed.

11.4 RLS Validation

Both regional roles were tested using Power BI Desktop's View As feature.

The report was checked to ensure that the regional filters affected the dashboard results.

11.5 Refresh Validation

The local Excel workbook was used as the source for a Power BI Desktop refresh test.

The refresh process was checked to ensure that the queries and report continued to work.

12. BUSINESS QUESTIONS ADDRESSED

The dashboards support investigation of several business questions.

Sales Analysis

- What is the overall sales performance?
- How do sales change over time?
- Which product categories contribute the most sales?
- Which markets generate higher sales?
- Which products perform best?

Customer Analysis

- How many unique customers are present?
- Which customer segments contribute the most sales?
- Which customers generate the highest sales?
- How does customer activity vary across regions?

Supply Chain Analysis

- How many orders are flagged with delivery risk?
- Which regions have higher late-delivery risk?
- Which shipping modes are used most frequently?
- How do actual shipping days compare with scheduled shipping days?

Executive Analysis

- What is the overall business performance?
- Which regions contribute most to sales?
- What are the important operational indicators?
- Which areas may require further investigation?

13. CHALLENGES FACED AND SOLUTIONS

During project development, several technical challenges were encountered.

Challenge	Solution
Date and time conversion issues	Corrected date data types using Power Query and prepared a date-only field.
Ambiguous relationship paths	Reviewed the model and disabled loading for the original staging query where appropriate.
Region slicer relationship issue	Used the working FactSales region field for the slicer.
KPI displaying 0.55 instead of 55%	Changed the measure format to Percentage.
Top 10 customer chart filtering and sorting difficulties	Configured a Top N filter based on Total Sales and reviewed the visual configuration.
Understanding Row-Level Security	Created regional roles and tested them using View As.
Data refresh understanding	Established a manual refresh workflow using the local Excel source workbook.

14. PROJECT OUTCOMES

The project resulted in the development of an interactive Power BI reporting solution.

The major outcomes include:

1. A consolidated reporting model using sales and supply chain data.
2. Data preparation and transformation using Power Query.
3. A fact-and-dimension data model.
4. Reusable DAX measures for important business KPIs.
5. Four interactive dashboard pages.
6. Regional data access restrictions using Row-Level Security.
7. A repeatable manual data refresh workflow.
8. Validation of important measures and report interactions.

The project demonstrates practical application of data analytics and business intelligence concepts.

15. LIMITATIONS AND FUTURE ENHANCEMENTS

Although the current implementation provides important analytical capabilities, several enhancements can be considered.

15.1 Current Limitations

- The dataset does not include complete HR information.
- Complete financial statement reporting is not supported by the selected dataset.
- Data refresh is performed manually.
- Static RLS roles are used instead of dynamic user-based security.
- Automated ETL pipelines are not implemented.
- Cloud-based report deployment and user provisioning are outside the current implementation.

15.2 Future Enhancements

1. Implement dynamic Row-Level Security.
2. Integrate additional HR and finance datasets.
3. Develop automated data ingestion pipelines using Azure Data Factory or scheduled SQL jobs.
4. Implement scheduled refresh using an appropriate cloud or gateway configuration.
5. Improve time-series analysis using Year-Month reporting.
6. Introduce advanced customer segmentation.
7. Develop predictive models for sales forecasting and delivery-risk analysis.
8. Extend the reporting solution with additional business KPIs.

16. CONCLUSION

The Enterprise Business Intelligence Suite demonstrates the development of an end-to-end Power BI reporting solution using a real-world supply chain dataset.

The project involved importing and transforming data, designing a star-schema data model, developing DAX measures, building interactive dashboards and implementing regional Row-Level Security.

Four dashboards were created to analyze executive performance, sales, supply chain operations and customer behaviour.

The implementation also demonstrates data validation and a repeatable manual refresh process using a local Excel workbook.

Overall, the project provided practical experience in data preparation, data modelling, business intelligence, analytical reporting and data security using Microsoft Power BI.

The solution serves as a foundation for further development into a more automated and scalable enterprise Business Intelligence platform.